FEDERAL SPECIFICATION

MOTOR, DIRECT CURRENT (INTEGRAL HORSEPOWER, 200 HP AND SMALLER)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers integral horsepower motors (200 horsepower and smaller) for use on nominal 120 or 240 volts direct current (d.c.) systems in nontactical equipment.

1.2 Types and classes. Motors furnished under this specification shall be of one type (integral horsepower) and of the following classes, as specified (see 6.1):

Class 1—Shunt wound.
Class 2—Series wound.
Class 3—Compound wound.

1.2.2 *Styles.* Motors furnished under this specification shall be of the following styles, as specified (6.1).

Styles:

A1—General purpose
A2—Drip proof
A3—Splashproof
A4—Guarded
A5—Drip proof, fully guarded
B1—Nonventilated
B2—Fan-cooled
B3—Waterproof
B4—Explosion proof
B5—Dust-ignition proof

Note 1. Style A motors are variations of the open motor.
Style B motors are variations of the totally-enclosed motor.

Note 2. For definitions of the various electrical terms used throughout this specification, reference should be made to ASA C42.10.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification:

Federal Specifications:

FF-B-171—Bearings, Ball, Annular (General Purpose).
FF-B-185—Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187—Bearing, Roller, Tapered.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-V-173—Varnish, Moisture-and-Fungus-Resistant (for the Treatment of Communications, Electronic, and Associated Electrical Equipment).
MIL-V-1137—Varnish, Electrical-Insulating (for Electro-motive Equipment).
MIL-I-11748—Interference Reduction for Electrical and Electronic Equipment.
MIL-E-16298—Electric Machines Having Rotating Parts and Associated Repair Parts; Packaging of.

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Underwriters' Laboratories, Inc. UL Standards:

UL-674(a)—Motors and Generators, Electric, for Use in Hazardous Locations, Class II, Groups F and G.
UL-674(b)—Motors and Generators, Electric, for Use in Hazardous Locations, Class I Groups C and D (Part I—Integral Horsepower).

(Applications for copies should be addressed to Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y., 11749; 207 East Ohio Street, Chicago, Ill., 60611; 1655 Scott Blvd., Santa Clara, Calif., 95050.)

American Standards Association, Inc., ASA Standards:

C1—National Eelctrical Code.
C6.1—Terminal Markings for Electrical Apparatus.
C42.10—Definitions of Electrical Terms: Rotating Machinery.
C50.4—Direct-Current Commutating Machines in General.
C50.8—Dimensions for Motors and Generators.
S1.2—Physical Measurement of Sound, Method for.

(Application for copies should be addressed to American Standards Association, Inc., 10 East 40th Street, New York, N. Y., 10016.)

3 REQUIREMENTS

3.1 Material. All material used in the construction of motors shall be of the quality best suited to the purpose intended. The material used in the construction of any design of motor in all cases shall be consistent with the strength required for safety and reliability

3.2 Duty. The duty of motors shall be continuous or short time, as specified (see 6.2).

3.2.1 *Continuous duty.* Continuous duty motors shall be rated at the load in horsepower that can be carried continuously at the ambient temperature specified without exceeding allowable temperature rises as specified in ASA C50.4.

3.2.2 *Short time duty.* Short time duty motors shall be rated at the horsepower load which, starting at the ambient temperature, can be carried constantly for the period specified without exceeding allowable temperature rises specified in ASA C50.4.

3.3 Voltage ratings. Motors shall be furnished with voltage ratings of 120 or 240 volts, as specified. When motors rated at voltages other than the above are required, the voltages shall be as specified in the invitation for bids (see 6.1). For motors of 15 horsepower or larger, 240 volts is considered the preferred standard.

3.4 Horsepower and speed ratings. The horsepower and speed ratings for d.c. inte-

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gral horsepower motors rated at 120 and 240 volts shall be in accordance with ASA C50.4 and as specified in the invitation for bids (see 6.1).

3.5 Overspeed. Direct current motors having the speed ratings specified in ASA C50.4 shall be so constructed that, in any emergency, they will withstand an overspeed of 25 percent above the highest rated speed without mechanical injury. Series wound motors must have the maximum safe operating speed stamped on the nameplate.

3.6 Ambient temperature. Motors shall be designed for the following ambient temperatures, as specified (see 6.1):

40°C. 50°C. Special

3.6.1 40°*C. ambient.* In general and except where otherwise specified, the rating of the equipment shall be based upon an ambient temperature of 40°C. for those installations where the maximum normal operating temperature of the surrounding atmosphere or other cooling medium is 40° C. or less.

3.6.2 50°*C. ambient.* In general and except where otherwise specified, the rating of the equipment shall be based upon an ambient temperature of 50°C. for those installations where the maximum normal operating temperature of the surrounding atmosphere or other cooling medium is 40°C. but not higher than 50°C.

3.6.3 *Special ambient.* Where the maximum normal operating temperature of the surrounding atmosphere or other cooling medium is in excess of 50°C., the rating of the equipment shall be based upon the special ambient temperature, as specified.

3.7 Electrical insulation.

3.7.1 *Systems.* The electrical insulation systems shall conform to ASA C50.4, class A, B, F, or H as specified (see 6.1).

3.7.2 *Material.* The class of insulation material used in all components of an electrical system shall conform to one of the following:

3.7.2.1 *Class* 105. Materials or combinations of materials such as cotton, silk, and paper when suitably impregnated or coated or when immersed in a dielectric liquid such as oil. Other materials or combinations of materials may be included in this class, if by experience or accepted tests they can be shown to be capable of operation at 105°C.

3.7.2.2 *Class* 130. Materials or combinations of materials such as mica, glass, fiber, asbestos, etc., with suitable bonding substances. Other materials or combinations of materials, not necessarily inorganic, may be included in this class, if by experience or accepted tests, they can be shown to be capable of operation at 130°C.

3.7.2.3 *Class* 155. Materials or combinations of materials such as mica, glass, fiber, asbestos, etc., with suitable bonding substances. Other materials or combination, of materials, not necessarily inorganic, may be included in this class, if by experience or accepted tests they can be shown to be capable of operation at 155°C.

3.7.2.4 *Class* 180. Materials or combinations of materials such as silicone elastomer, mica, glass, fiber, asbestos, etc., with suitable bonding substances such as appropriate silicon resins. Other materials or combinations of materials may be included in this class, if by experience or accepted tests they can be shown to be capable of operation at 180°C.

Note: 3 An insulation is considered to be impregnated when a suitable substance replaces the air between its fibers, even if this substance does not completely fill the spaces between the insulated conductors. The impregnating substances, in order to be considered suitable, must have good insulating properties; must entirely cover the fibers and render them adherent to each other and to the conductor; must not produce interstices within itself as a consequence of evaporation of the solvent or through any other cause; must not flow during the

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operation of the machine at full working load or at the temperature limit specified; and must not unduly deteriorate under prolonged action of heat.

3.7.2.5 *Mixed insulation.* Unless otherwise specified in the invitation for bids, nothing in the foregoing shall be construed as limiting all the insulation on a given motor to a single class. Class 130 insulation may, if desired, be used on a motor where class 105 insulation is specified or approved, and the temperature limits applicable to the particular class of insulation used shall be met in every case.

3.8 Commutators.

3.8.1 *General.* Commutators shall be built up of best quality hard-drawn, hard-rolled, drop-forged copper segments or other suitable metals insulated from each other by mica, pressed mica plate, or other suitable material, and from the shell by mica, pressed mica plate, molded insulation, or other suitable material. The commutator shall be secured rigidly to the shaft or spider by keying or other suitable methods which will positively insure maintenance of alignment with respect to the rotor core slots.

3.8.2 *Commutator segments.* The commutator segments shall be securely retained in a manner that will best prevent their relative displacement as a result of centrifugal forces and the stresses imposed by repeated expansion and contraction in service. The segments, when worn to the full extent of the wearing depth specified hereunder, shall be of ample section to operate satisfactorily. The wearing depth, measured on the radius, for commutators up to and including 6-inch diameters, shall be not less than 8 percent of the radius and in no case less than 1/8 inch. For commutators more than 6 inches in diameter, the wearing depth, measured on the radius, shall be not less than 5/16 inch.

3.8.3 *Connections.* The connections to the armature windings shall be efficiently soldered, brazed, welded, or by other suitable means, and if separate risers are used, they shall be mechanically and electrically connected to the commutator bar in a suitable manner.

3.8.4 *Sealing.* After the assembly of the commutator, all crevices or joints at each end between the bars and the retaining flanges shall be completely filled up and sealed over with a high-grade, flexible, baking, insulating varnish in a manner that will prevent the entrance of moisture, oil, carbon, or copper dust, or other deleterious substances except at points where the fill would interfere with the function. Likewise, the short creepage paths from the bare copper bars at each end to the bare metal flanges shall be given a heavy durable coating of flexible baked insulating varnish as a protection against short-circuits and grounds at these points.

3.8.5 *Undercutting.* The insulation between bars shall be undercut. The undercutting shall be accomplished by removing all insulation between bars to form a groove not to exceed 3/64 inch in depth.

3.8.6 *Curing.* All commutators shall be cured at higher than operation temperatures and shall be of such thoroughly solid construction throughout that they will hold their shape in service and obviate the necessity of frequently dressing the commutators on account of high bars or flat spots.

3.6 Brush holders.

3.9.1 The brush holders for reversible motors shall be of a type which will permit satisfactory operation with the motor running in either direction. They shall be protected by the motor and end shield and provided with means for shifting the brushes. On 7-1/2-horsepower and larger motors, the brush rigging shall be mounted on a rocker ring which shall fit into an accurately machined groove in the motor end shield or rigidly mounted to the end frame.

3.9.2 Brush holders shall be readily accessible for adjustment and renewal of

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brushes and springs. The springs shall not be depended upon to carry current.

3.9.3 The brush holders and rigging on 7-1/2-horsepower and larger motors shall be constructed to permit the proper staggering of the brushes to insure even wear of the commutator.

3.10 Brushes. Brushes shall be of sufficient capacity to prevent undue heating when the motor is carrying normal rated load under the most unfavorable service conditions. If the brush slides in the holder, it shall be connected to the brush holder by a flexible copper connector or "pigtail".

3.11 Winding connections and terminal leads.

3.11.1 *Electrical connectors.* A high temperature tin-lead-silver solder, pure tin solder, spelter solder, copper welding, or solderless clamp or crimp type connectors shall be employed in the making of all electrical connections in class 130, 155, or 180 insulated windings. If desired, a tin-lead solder may be used where the windings are class 105 insulated throughout. In motors employing either class 105, 130, 155, or 180 insulation, the welding together of stator connections by burning off the twisted ends is permissible in lieu of soldered joints.

3.11.2 *Securing of connections.* All electrical connections may be made by soldering, bolting or other suitable means which will give good electrical contact. Parts shall be so proportioned that no injurious heating will occur under the most unfavorable service conditions. Such connections shall be provided with any locking devices needed to prevent them from becoming loosened by vibration.

3.11.3 *Terminal leads.* All motors shall be provided with suitable and accessible leads for line connection. Leads shall be suitably secured to the winding and shall terminate in a suitable terminal box unless the motor is specified for use on equipment where other construction is suitable. Leads shall be made of flexible, stranded, insulated wire.

3.12 Terminal markings. All motor terminals shall be legibly and permanently marked in accordance with ASA C6.1 when more than two are brought out from the motor frame. When terminal markings are not covered by ASA C6.1 a terminal marking diagram shall be permanently affixed to the motor.

3.13 Bearings and shafts. Bearings and shafts shall be designed to prevent the excape of the lubricant, when applicable. A shaft extension, with keyway (including key) or flat, shall be provided for the driven apparatus. Ball and roller bearings shall conform to FF-B-171, FF-B-185, or FF-B-187, as applicable.

3.13.1 *Bearings, horizontal motors.* Horizontal motors may be furnished with either sleeve, roller or ball bearings, unless one particular type is specified in the invitation for bids. Where sleeve bearings are used, ample bearing surface shall be provided and the bearing shall be efficiently lubricated.

3.13.2 *Bearings, vertical motors.*

3.13.2.1 Vertical motors may be furnished with either sleeve, roller or ball bearings, unless one particular type is specified in the invitation for bids.

3.13.2.2 Where sleeve bearings are used, ample bearing surface shall be provided and the bearing surface shall be efficiently lubricated. One of the sleeve bearings, or a separate device, shall have adequate thrust capacity to support the weight of the motor rotor and, where directly connected to the driven unit, the thrust transmitted from the connected unit, as specified in the invitation for bids.

3.13.2.3 Where ball or roller bearings are used, the bearing shall be efficiently lubricated and shall have adequate thrust capacity to support the weight of the motor rotor and, where directly connected to the

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driven unit, the thrust transmitted from the connected unit, as specified in the invitation for bids.

3.13.2.4 In the case of vertical motors, the requirements of 3.13 shall apply with special emphasis.

3.14 **Feet.** When motors intended for foot mounting are purchased unattached to any equipment, the motor frame shall be provided with feet, practically true and drilled or slotted for holddown bolts.

3.15 **General construction.** End shields shall be held in place by close-fitting rabbit joints with not less than two machine screws or bolts of suitable size and strength. When so specified, the design of the end shields on horizontally mounted motors shall provide for, and the accuracy of machining shall be such as to permit the rotation of the bearings through 90° to 180° in either direction, to allow for the mounting of the motor on the ceiling or underside of the apparatus. Machine screws or bolts shall be either hexagon-headed or furnished with screwdriver slots; those necessary to be removed for maintenance shall be screwed into metal.

3.16 **Dimensions.** Unless otherwise specified, all dimensions and tolerances shall conform to ASA C50.8.

3.17 **Electrical performance.**

3.17.1 *Insulation resistance.* The insulation resistance, when corrected to 25°C., shall be not less than the following:

Armature circuit (class 105 insulated) 12 megohms
Armature circuit (class 130, 155, and 180 insulated) 25 megohms
Field circuit (class 105 insulated) 25 megohms
Field circuit (class 130, 155, and 180 insulated) 50 megohms

3.17.2 *Dielectric strength.* The dielectric strength shall be in accordance with ASA C50.4.

3.17.3 Unless otherwise specified, the following requirements shall conform to ASA C50.4:

(a) Speed regulation
(b) Temperature rise*
(c) Overload and commutation

* For 50°C. ambient temperature, the maximum observable temperature rise shall be 10°C. less than the values specified for 40°C. ambient temperature.

3.18 **Balance.** Motors shall run well balanced at all speeds and loads within the maximum service range. All motors and separable fans shall be accurately balanced by drilling or other suitable means. Uninsulated balancing wedges may be used in armature slots provided that not more than two wedges, separated at least by 1/8 inch, are fitted in any slot and the length of each wedge does not exceed 30 percent of the length of the rotor core. Maximum peak-to-peak displacement shall conform to the requirements of ASA C50.4.

3.19 **Audible noise.** Motors shall operate without abnormal audible noise at all loads and speeds within the service range (see 4.4.2).

3.20 **Explosion-proof motors.**

3.20.1 Standard of Underwriters' Laboratories, Inc., UL-674 (a) or (b) for electric motors and generators for use in hazardous locations shall apply to all purchases of dust-ignition-proof and explosion-proof motors.

3.20.2 The contractor shall submit evidence that the dust-ignition-proof and explosion-proof motors he proposes to supply under this specification conform to the standards of UL, as regards fire and casualty hazards. The listing of UL for dust-ignition-proof and explosion-proof motors will be accepted as evidence that the manufacturer can supply such motors conforming with this requirement. Such listings shall be cited in the bid for each type and size of dust-ignition-proof and explosion-proof

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motor offered, or, in lieu thereof, the bid must be accompanied by a certified test report from an independent testing laboratory stating that tests, conducted in substantial accordance with tests outlined in UL Standard for Electric Motors and Generators for use in Hazardous Locations show that the sizes and types of motors offered conform to UL standard.

3.20.3 Compliance of dust-ignition-proof and explosion-proof motors with all requirements of UL Standard for Electric Motors and Generators for Use in Hazardous Locations for the types and sizes of motors offered is necessary. The labels of UL for electric motors for hazardous locations may be accepted as evidence of such compliance, or compliance may be determined by laboratory tests, but this does not absolve the bidder from complete compliance with the additional requirements of the specification in order to secure the acceptance of his dust-ignition-proof and explosion-proof motors.

3.21 Radio-interference suppression. When specified in the invitation for bids, all motors shall be equipped for radio-interference suppression in accordance with MIL-I-11748, class III a, unless procurement is under an end equipment specification in which a less stringent end equipment suppression requirement may apply. Conformance to MIL-I-11748 requirements will be determined by authorized representatives of the contracting officer using Government test equipment. The testing agency will furnish the contracting officer a report of the model tested, including details of the suppression system applicable for production units and recommendations for quality control of production. Upon approval by the contracting officer and provided all other requirements of this specification are met, the contractor shall examine each production unit to determine that it conforms to the report of the contracting officer and the approved suppression system of the first unit of production.

3.22 Fungus control of electrical components. When specified in the invitation for bids, motors shall be treated to resist growth as follows:

3.22.1 *Motor windings.* The motor windings shall be given 2 impregnating coats of varnish conforming to type M, grade BB or grade CB, and one sealer coat conforming to type M, grade CA, of MIL-V-1137. The coats shall be applied by one or more of the following methods of impregnation:

Vacuum-pressure
Immersion
Centrifugal
Pulsating-pressure
Built-up
Brushing or spraying (for sealer coat only)

(a) The vacuum-pressure method is the preferred method. It shall consist of an arrangement for heating the windings in a vacuum chamber, at the temperature and for the period of time necessary to provide for the evacuation of all entrapped moisture and air, and then, without breaking the vacuum, allowing the liquid insulation (maintained at the proper temperature) to flow into the windings under graduated pressure. This process shall be followed by necessary draining and baking. The preheating of the windings may be omitted in the case of windings where the conductor insulation does not employ material such as cotton or asbestos and is of synthetic resinous material.

(b) The immersion method shall consist of first heating the windings to the temperature and for the period of time necessary to thoroughly dry them out, and then completely immersing the windings in insulating material for the period of time necessary for thorough penetration. The baking may be omitted in the case of windings where the conduc-

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tor insulation does not employ hygroscopic material such as cotton or asbestos and is of synthetic resinous material. This method will not be given approval if less than two dips and bakes have been applied. All dips and bakes on the individual coils during the process of manufacture as well as the final dipping and baking may be used in determining the total number of dips and bakes.

(c) The centrifugal method is a pressure-impregnation method, the pressure being the effect of centrifugal force due to the rotation of the chamber containing the insulating material and the windings to be impregnated.

(d) The pulsating-pressure method is a pressure-impregnation method consisting of repeated applications of pressure for suitable periods of time alternating with zero pressure.

(e) The built-up method shall consist of the impregnation and curing of the insulation at each step in the building up of the windings.

(f) The brushing or spraying method is not considered as an adequate method, in itself, of applying insulating material to electrical windings. The use of this method shall be limited to the sealer (final) coat of varnish applied to windings previously impregnated by one of the other methods outlined in the foregoing paragraphs.

3.22.2 *Circuit elements.* When specified in the invitation for bids, circuit elements, such as cable, wire, terminal blocks, condensers, coils, etc., which operate normally at a temperature of 200°F. or lower, shall be coated with a fungus-resistant varnish conforming to MIL-V-173, except those elements which are fabricated of materials inherently inert to fungus attack or which have been rendered resistant to fungus attack by approved treatments.

3.23 Drain plugs. The frames of all waterproof motors shall be provided with drain plugs or check valves at accessible points and so located that they will drain all pockets in the motor casing.

3.24 Enclosing covers. Enclosing covers, where used, shall be attached to the motor frame or end shields in a suitable manner. They shall be readily removable. In the case of waterproof motors, the contact surfaces between the enclosing covers and the motor frame shall be free from fins, burrs, and other imperfections detrimental to waterproofness and shall be provided with suitable gaskets, suitably secured and treated on the exposed surface to prevent sticking.

3.25 Finishing. The motor enclosure shall be neatly finished and suitably protected against corrosion.

3.26 Suitability. In the case of motor-driven apparatus where the motor is furnished as part of a complete operating unit, the motor shall possess those particular characteristics, as specified in the invitation for bids, which render it inherently suitable for and adapted to the purpose for which it is being used; and shall, also, comply with all parts of this specification which are not in conflict with these particular requirements. The capacity of the motor shall be sufficient to safeguard it against overload when driving the specified equipment at normal output under the most unfavorable service conditions to which the equipment may be subjected.

3.27 Nameplate data. The nameplate data shall conform to ASA C50.4.

3.28 Plans with bids.

3.28.1 *General.* In the case of motors to be purchased solely as motors (not a part of other apparatus), each bidder shall submit with his bid, sufficient information in duplicate to enable the activity concerned to obtain a clear understanding of the apparatus offered before making recommendation of award of contract. This informa-

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tion may be in the form of blueprint plans, photographs, or typewritten or printed descriptive specification, in such arrangement as the bidder may prefer.

3.28.2 *Information required.* The information contained in any plans with bids shall clearly show the following:

Outline, overall and principal dimensions, weights.

Complete rating (voltage, horsepower, speed, full-load current).

In addition, when specifically called for in the invitation for bids, the guaranteed performance, including motor efficiency, at 3/4 and full rated load, shall be indicated; also 1/2 load and idling, when required.

3.29 Identification marking. The motor shall be marked for identification in accordance with MIL-STD-130.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling. If any department or independent establishment requires laboratory tests, one sample of each type and rating of motor purchased, shall be supplied to the testing laboratory specified by the contracting officer.

4.3 Inspection. Each motor shall be subjected to a thorough examination to ascertain that the material, workmanship, and design (type, degree of enclosures, lubrication, etc.) are in accordance with the requirements of this specification.

4.4 Tests.

4.4.1 *General.* Unless otherwise specified, motors selected in accordance with 4.2 shall be subjected to the tests necessary to determine compliance with the requirements of this specification. The tests shall be conducted as specified in ASA C50.4.

4.4.2 *Audible noise.* Unless otherwise specified, audible noise measurements shall be made and determined in accordance with ASA S1.2.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the various levels of packaging and packing protection for supplies and equipment shall be as specified in Fed. Std. No. 102.)

5.1 Preservation and packaging, level A and C. The motors shall be preserved and packaged in accordance with level A or C of MIL-E-16298, as specified.

5.2 Packing, level A, B, and C. The motors shall be packed in accordance with level A, B, or C of MIL-E-16298, as specified.

5.3 Marking.

5.3.1 *Civil agencies.* Unless otherwise specified in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Class and style of motor (1.2.1 and 1.2.2).

(*c*) Duty (continuous, short time, etc.) (3.2).

(*d*) Voltage (3.3).

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(*e*) Horsepower and speed ratings (3.4).
(*f*) Ambient temperature (3.6).
(*g*) Insulation class (3.7).
(*h*) Bearing type (3.13).
(*i*) Type and number of shaft extensions, including details when other than standard shaft extensions are required (3.13).
(*j*) Type of mounting (horizontal or vertical).
(*k*) Vertical thrust requirements (3.13.2.2 and 3.13.2.3).
(*l*) Method of mounting (feet, etc.) (3.14).
(*m*) Dimensions, if different from those specified (3.16).
(*n*) Performance characteristics, if different from those specified (3.17.3).
(*o*) Radio-interference suppression, if required (3.21).
(*p*) Fungus-preventative treatment, if required (3.22).
(*q*) Suitability when supplied as a part of a complete operating unit (3.26).
(*r*) Performance guarantee, if desired (3.28.2).
(*s*) Spare parts and special tools, if any.
(*t*) Inspection responsibility, if other than specified (4.1).
(*u*) Laboratory tests, if desired (4.2).
(*v*) Tests, if other than specified (4.4).
(*w*) Audible noise measurement, if not required (4.4.2).
(*x*) Preservation, packaging, and packing required (4.4.2).
(*y*) Marking, if other than specified (5.1 and 5.2).

6.2 Proper selection of apparatus. Extreme care should be used in the proper selection of apparatus in order that successful operation and good service will result. When in doubt, the exact type of service should be specified. Where the apparatus is subjected to unusual risk, manufacturers should be consulted, especially where the apparatus is used under the following conditions.

(*a*) Exposed to chemical fumes (specify chemical).
(*b*) Operated in damp places.
(*c*) Operated at excess speed (specify maximum if possible).
(*d*) Exposed to combustible or explosive dust.
(*e*) Exposed to gritty or conducting dust.
(*f*) Exposed to lint.
(*g*) Exposed to steam.
(*h*) Operated in poorly ventilated spaces.
(*i*) Operated in pits, or where entirely enclosed in boxes.
(*j*) Exposed to inflammable or explosive gases or vapors.
(*k*) Exposed to temperatures below 10° C.
(*l*) Exposed to temperatures above 40° C.
(*m*) Exposed to oil vapor.
(*n*) Exposed to salt air.
(*o*) Exposed to abnormal shock or vibration from external sources.
(*p*) Where the departure from rated voltage or frequency is excessive (specify limits).
(*q*) Exposed to liquid sprays (specify liquid).
(*r*) Operated at high elevation (specify elevation).
(*s*) Momentarily submerged in water.
(*t*) Operated in air current, as when driving fans.

It is suggested that the text of 1.2 through 1.2.2 be scanned closely and the proper choice under each individual paragraph be made so that the summation of all the choices or options will give an adequate description of the desired motor.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Motors, electric, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.

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Motor:

Motors, electric, not otherwise indexed. Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY INTEREST:

None

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